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Project Report On

“Marketing Mix Modeling: A Universal Framework for FMCG Sales Attribution”

BY

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1RUB24MBA0012

FOR FULFILLMENT OF THE REQUIREMENT FOR THE
AWARD OF THE DEGREE
OF
MASTER OF BUSINESS ADMINISTRATION

UNDER THE GUIDANCE OF

Mrs. Suma M N

Assistant Professor

SCHOOL OF BUSINESS
RV UNIVERSITY

CONSENT LETTER FOR PROJECT SUPERVISION

To

The Dean/Program Head/Course Co-Ordinator
School of Business
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Subject: Consent letter to be a Dissertation Supervisor for Mr. Arun KN

Program Name: MBA – Artificial Intelligence and Data Science

Subject: Dissertation project.

Topic: Marketing Mix Modeling: A Universal Framework for FMCG Sales Attribution

With reference to the above-mentioned subject, I wish to inform you that I am willing to accept Mr. Arun KN 1RUB24MBA0012 under my guidance to work on the topic “Marketing Mix Modeling: A Universal Framework for FMCG Sales Attribution” towards the fulfillment of the requirement to complete a 6-credit Dissertation at RV University. I will guide them for the entire duration of the project, and I will supervise their work throughout the process.

Thank you

Yours Sincerely

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Signature of the Project guide:

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Signature:

Date: 19/ 02/2026

Place: Bangalore

DECLARATION BY THE STUDENT

Date: 16/04/2026

DECLARATION

I Arun KN, 1RUB24MBA0012, fourth-semester student of MBA declare that the Dissertation Report titled, “Marketing Mix Modeling: A Universal Framework for FMCG Sales Attribution “ submitted for the degree of Master of Business Administration Artificial Intelligence and Data science to the RV University, is my original work, conducted under the supervision of my guide Mrs.Suma M N. I also wish to inform you that no part of the project work has been submitted for a degree or examination at any University. I am aware of and understand the university’s policy on plagiarism and I certify that this report is my work, except where indicated by reference.

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CERTIFICATE FROM THE SUPERVISOR

CERTIFICATE

This is to certify that **Mr.Arun KN** bearing **USN 1RUB24MBA0012** is a Bonafide student of Master of Business Administration of the RV University (Batch 2024), Bengaluru.

We confirm that the Dissertation Report titled “**Marketing Mix Modeling: A Universal Framework for FMCG Sales Attribution**” is authored by him under the guidance of Suma M N, in partial fulfillment of the requirements for the award of the Master of Business Administration of RV University, Bengaluru, Karnataka.

I, further certify that this work has not formed the basis for the award of any Degree, Diploma, Associateship, Fellowship, or other Titles in this University or any other University or institution of Higher Learning.

Signature

Name of the Supervisor: Mrs. Suma M N

Designation: Assistant Professor

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Arun KN

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CHAXPTER 1: INTRODUCTION AND BUSINESS CONTEXT

1.1 Industry Background

The Fast-Moving Consumer Goods (FMCG) industry is one of the most competitive and marketing-intensive sectors in the global economy. Companies operating in this space spend billions of dollars every year across a wide mix of media channels: television, digital platforms, radio, influencer marketing: with the expectation that these investments will translate into measurable sales outcomes. The challenge, however, is that most organizations struggle to precisely quantify how much of their revenue can actually be attributed to each of these channels. This is not a new problem, but it has become far more complex in the current environment where the number of touchpoints a consumer encounters before making a purchase decision has multiplied dramatically

India's FMCG sector, in particular, has witnessed an aggressive shift in how brands communicate with consumers. Traditionally dominated by mass media like television and print, the channel landscape has now expanded to include Facebook, YouTube, TikTok, and influencer-led content. With this expansion comes a fundamental measurement problem: when multiple channels are running simultaneously, how does a brand manager know which one is actually driving units sold? This is the problem that Marketing Mix Modeling (MMM) was designed to solve, and it is the central focus of this dissertation

1.2 Business Problem and Research Gap

Despite the availability of advanced analytics tools, a significant portion of FMCG brands in India still rely on either intuition or very basic attribution methods when making media budget decisions. Platforms like Google and Facebook offer their own attribution reports, but these are inherently biased toward their own channel. What is missing is an independent, model-driven framework that ingests data from all channels

simultaneously and outputs a clear, mathematically grounded contribution percentage for each one

Existing MMM solutions tend to fall into two camps: expensive proprietary tools used by large multinationals, or academic Bayesian frameworks that require significant statistical expertise to operate. What is missing for mid-sized FMCG brands is a flexible, universal framework that any analyst can plug a cleaned dataset into and extract reliable channel contributions from. This dissertation attempts to fill that gap by building and demonstrating exactly such a system

Additionally, there is a well-documented challenge in MMM research around model selection: different model architectures produce very different contribution estimates for the same dataset. A linear regression might attribute 49% of sales to one channel while a Bayesian model puts the same channel at 6%. Understanding which model to trust, and why, is an under-discussed but practically critical question. This project addresses that directly by running seven different model architectures on the same dataset and comparing their outputs

1.3 Research Objectives

The primary objectives of this dissertation are:

1. To design and implement a universal data cleaning and preprocessing pipeline capable of handling real-world FMCG sales data with missing values, corrupted entries, and inconsistent formats
2. To build an AI-powered autonomous machine learning modeling engine that selects, trains, and iteratively optimizes regression models for sales prediction without manual intervention

3. To develop a Marketing Mix Model that incorporates Adstock transformations and Hill saturation curves to account for media carryover and diminishing returns effects
4. To compare the channel contribution outputs of multiple model architectures: including Linear Regression, Ridge, Lasso, Random Forest, XGBoost, OLS, and Bayesian (PyMC): and derive managerially actionable insights from the comparison
5. To validate the entire three-stage framework using synthetic FMCG datasets and demonstrate its applicability as a reusable analytical tool for media budget optimization

1.4 Research Questions

The following research questions guide this study:

- How can raw, messy FMCG sales data be systematically transformed into a model-ready format using an automated pipeline?
- Which machine learning model architecture best predicts FMCG unit sales, and what features drive that prediction?
- How much of total FMCG sales can be attributed to paid media channels (TV, Facebook, YouTube, Radio, TikTok, Influencer) versus base sales after controlling for Adstock and saturation effects?
- Why do different model architectures produce significantly different channel contribution estimates, and which should a brand manager rely on?

1.5 Scope of the Study

This study is scoped around the development and validation of a three-module analytical framework for FMCG marketing analytics. The first module covers data

ingestion and cleaning. The second covers predictive sales modeling using an autonomous ML agent. The third covers Marketing Mix Modeling with multi-model comparison

The framework has been validated using synthetic datasets purpose-built to reflect real FMCG data characteristics: weekly sales records, promotional flags, stock availability, and spend data across six media channels. The datasets were designed to introduce realistic noise and data quality issues so the pipeline's cleaning capabilities could be genuinely tested

The scope does not extend to real-time deployment or integration with live data systems. It also does not cover attribution modeling for e-commerce platforms or digital-first brands, though the framework's architecture is flexible enough to accommodate these with appropriate modifications. The study is primarily quantitative in nature and relies on secondary synthetic data rather than primary survey-based research

1.6 Significance of the Study

From an academic standpoint, this project contributes a practical, working implementation of MMM that bridges the gap between theoretical frameworks discussed in literature and the operational reality of analysts working with imperfect data. Most academic papers on MMM present idealized datasets and single-model implementations. This project explicitly confronts data messiness, model selection uncertainty, and multi-architecture comparison in a way that is rare in student-level research

From a managerial standpoint, the framework directly addresses one of the most common pain points in FMCG marketing: the inability to justify media budget allocations with hard numbers. A brand manager using this system would be able to

take any historical sales dataset, run it through the pipeline, and within a reasonable compute time receive a clear breakdown of how much each media channel contributed to revenue. This removes guesswork from one of the most expensive decisions a marketing team makes

The use of Lasso regularization as the best-performing model (Test R^2 of 0.9726, MAPE of 3.07%) also carries a specific practical implication: it naturally zeroes out weak or irrelevant channels, acting as an in-built budget efficiency filter. Combined with the Bayesian model's uncertainty quantification, this framework gives decision-makers both a point estimate and a range of confidence for every channel's contribution

1.7 Structure of the Report

This dissertation is organized across five chapters. Chapter 1 (this chapter) establishes the business context, defines the problem, and outlines the objectives and scope of the study. Chapter 2 reviews the relevant literature on Marketing Mix Modeling, Adstock transformations, Bayesian versus frequentist modeling approaches, and the role of machine learning in marketing analytics. Chapter 3 describes the research methodology in detail: covering the architecture of all three framework modules, the datasets used, and the analytical techniques applied. Chapter 4 presents the results from all three stages: data cleaning outcomes, ML model performance, and MMM channel contribution findings: and discusses their implications. Chapter 5 concludes the study with key findings, practical recommendations for FMCG brand managers, limitations of the current framework, and directions for future research

CHAPTER 2: LITERATURE REVIEW AND CONCEPTUAL FRAMEWORK

2.1 The Origins of Marketing Mix Modeling

The concept of the marketing mix has its roots in the early work of Neil Borden (1964), who described the role of a marketing manager as a 'mixer of ingredients': someone who blends promotional spend, pricing, distribution, and product decisions to achieve a desired sales response. Jerome McCarthy (1964) later simplified this into the widely adopted 4Ps framework: Product, Price, Place, and Promotion. While these frameworks established the conceptual foundation for thinking about how marketing inputs drive business outcomes, they offered no quantitative method for isolating the effect of each individual input on sales

The need for such a quantitative method became urgent through the 1970s and 1980s as television advertising spend grew substantially and brand managers found themselves unable to justify budget allocation decisions with data. Early econometric approaches, drawing on work by researchers in marketing science, attempted to model the relationship between advertising spend and sales using regression analysis. However, these early models suffered from a critical limitation: they treated advertising as having an immediate, same-period effect on sales and ignored the reality that advertising continues to influence consumer behaviour for weeks after exposure. This created systematically underestimated ROI calculations and distorted channel attribution

Modern Marketing Mix Modeling (MMM) emerged as a structured solution to this problem. At its core, MMM is a statistical technique that uses historical data on media spend, pricing, promotions, and external factors to decompose total sales into contributions from each input variable. The output is a set of attribution percentages that tell a brand manager exactly how much of their revenue came from TV, digital,

radio, or base demand: information that directly informs budget planning and media strategy

2.2 Adstock Transformations: Theory, Justification, and Choice

One of the most significant conceptual advances in MMM was the introduction of the Adstock transformation by Simon Broadbent in his 1979 paper 'One Way TV Advertisements Work,' published in the Journal of the Market Research Society. Broadbent's central insight was that advertising exposure does not produce a single-period spike in awareness and then disappear: instead, it builds a stock of awareness in consumer memory that decays gradually over time. Each new exposure replenishes this stock, while the absence of advertising causes it to diminish at a rate determined by the medium, the creative, and the product category

The mathematical formulation Broadbent proposed is elegantly simple: the adstock at time t is equal to the advertising spend at time t plus a proportion of the adstock from the previous period. This decay proportion, denoted λ , is the carryover rate and lies between 0 and 1. A λ of 0 means there is no carryover: every week's advertising effect is independent. A λ close to 1 means the effect persists for a long time. The concept of half-life, borrowed from nuclear physics, has since become the standard way practitioners communicate adstock strength: a two-week half-life, for instance, means the incremental sales impact of an advertising burst takes two weeks to reduce by half (Broadbent, 1979; Fry, Broadbent & Dixon, 2000)

For FMCG brands specifically, empirical evidence across multiple studies suggests that adstock half-lives tend to fall between 2 and 5 weeks, with the average hovering around 25 weeks (Joseph, 2006). TV tends to exhibit the most predictable geometric decay, making it the channel for which simple adstock modeling is most reliable. Digital channels like Facebook and YouTube, by contrast, can exhibit more complex patterns: sometimes peaking in effect days after the spend rather than immediately, a

phenomenon captured by the delayed adstock variant of Broadbent's original formulation (Jin et al, 2017)

Three main approaches to modeling adstock carryover have been proposed in the literature and it is important to understand why geometric adstock was chosen for this study over the alternatives. The geometric decay model assumes a constant rate of decline each period and is computationally simple, well-understood, and has been validated in decades of practical MMM work. The Weibull distribution approach, used in Meta's Robyn framework, allows the decay rate to vary over time and can model initial buildup before decline: it is more flexible but requires more parameters and adds computational cost. Polynomial distributed lag models, proposed by Jastram (1955) and Palda (1965), use a finite number of lags with a polynomial restriction on their weights: they are less commonly used in modern practice

The geometric adstock was chosen for this framework because it strikes the right balance between theoretical soundness and operational simplicity. For a universal framework designed to work across datasets with varying sample sizes, the added parameter complexity of Weibull adstock introduces estimation instability: particularly problematic in datasets with limited weeks of data. Geometric adstock has a single tunable parameter (λ) that can be optimized per channel through grid search, making it tractable and interpretable. This reasoning aligns with the recommendation in Jin et al. (2017) who note that geometric adstock performs reliably across most practical MMM scenarios and is the appropriate default when the modeler does not have strong prior information about channel-specific decay shapes

2.3 Diminishing Returns and the Hill Saturation Function

The second major non-linearity that MMM must account for is diminishing returns to advertising spend. The economic intuition is straightforward: the first thousand rupees spent on Facebook advertising reaches a largely unaware audience and produces a meaningful sales lift; the ten-thousandth thousand rupees reaches an already-saturated

audience and produces a much smaller marginal return. If this effect is not modeled, MMM will systematically overestimate the value of high-spending channels and underestimate the value of under-invested ones

The S-curve and concave response functions have been discussed in advertising literature since Simon and Arndt (1980), who documented the shape of advertising response functions across product categories. Little (1979) proposed a general model for market response that accommodated both increasing and diminishing returns depending on spend levels. The modern MMM practice has largely converged on the Hill function, originally developed in biochemistry to model enzyme-substrate reactions, as the preferred approach for capturing saturation effects. As described in Jin et al. (2017), the Hill function maps advertising spend to a response value between 0 and 1, controlled by two parameters: the half-saturation point (the spend level at which half of maximum response is achieved) and the slope parameter (which determines how steeply the saturation curve rises)

The Hill function was chosen over simpler logistic or negative exponential saturation functions for this study because it offers greater flexibility in representing the full range of possible channel response shapes. Some channels exhibit a relatively flat, gradual saturation curve: meaning they remain efficient across a wide range of spend levels. Others saturate sharply, producing most of their incremental sales at low spend levels and very little beyond a threshold. The Hill function can represent both patterns through parameter adjustment, whereas a fixed-form logistic curve cannot. This is particularly important in a universal framework where the same model must handle channels with very different spending dynamics simultaneously

Importantly, the standard practice in the literature is to apply adstock transformation first and then the saturation transformation: which is the sequence followed in this study. This ordering is consistent with the recommendation in Jin et al. (2017): adstock captures the temporal carryover of spend, and then saturation is applied to the adstocked signal to capture the diminishing returns on the accumulated exposure rather than on

instantaneous spend. Reversing this order, as some practitioners do, tends to produce less interpretable and less stable parameter estimates

2.4 Frequentist Approaches to MMM: OLS, Ridge, and Lasso

The traditional workhorse of MMM has been Ordinary Least Squares (OLS) regression, which estimates channel coefficients by minimizing the sum of squared residuals between predicted and actual sales. OLS has the advantage of producing unbiased estimates when its assumptions are met: linearity, homoscedasticity, and independence of errors. In practice, however, MMM datasets routinely violate the most critical of these assumptions: multicollinearity. When multiple media channels are spending simultaneously over the same time period, their spend variables tend to be highly correlated with each other. This causes OLS coefficient estimates to become unstable: small changes in the data produce large swings in attributed channel contributions. The TikTok anomaly observed in this study's results, where OLS attributed 49.54% of sales to TikTok while Lasso attributed only 1.04%, is a direct consequence of this problem

Ridge regression (L2 regularization), proposed by Hoerl and Kennard (1970), addresses multicollinearity by adding a penalty term to the OLS objective function proportional to the square of each coefficient. This forces the model to shrink all coefficients toward zero simultaneously, distributing the attribution more evenly across correlated variables. Ridge is particularly useful in MMM when multiple channels have similar spending patterns and the analyst does not want to zero out any channel's contribution entirely. The limitation of Ridge is that it retains all variables in the model: even genuinely weak or irrelevant channels receive a non-zero attribution, which can overstate their importance

Lasso regression (L1 regularization), proposed by Tibshirani (1996), applies a penalty proportional to the absolute value of each coefficient. Unlike Ridge, this penalty creates a sparse solution: coefficients for weak or irrelevant variables are driven exactly to zero. In the context of MMM, this makes Lasso a natural variable selection tool: it effectively

tells the brand manager which channels have a real, statistically defensible contribution and which do not. The superior performance of Lasso in this study (MAPE of 3.07%, R^2 of 0.9726) relative to Ridge (MAPE 14.29%, R^2 of 0.3959) and OLS (MAPE 3.21%, R^2 of 0.9683) is consistent with findings in the literature that regularization, and L1 in particular, significantly improves MMM performance in the presence of multicollinear media spend data (Tibshirani, 1996; Zhou et al, 2021)

2.5 Bayesian MMM: Google's Framework and Its Advantages

The Bayesian approach to MMM was formally introduced to the practitioner community through the landmark paper by Jin, Wang, Sun, Chan, and Koehler (2017) from Google Research, titled 'Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects'. This paper proposed a unified probabilistic framework that incorporated both adstock carryover and Hill saturation directly into the model structure and estimated all parameters simultaneously using Markov Chain Monte Carlo (MCMC) sampling. The key theoretical advantage over frequentist approaches is that Bayesian inference treats model parameters as probability distributions rather than point estimates: meaning every channel attribution comes with a credible interval that quantifies uncertainty around the estimate.

This uncertainty quantification is practically valuable in ways that frequentist MMM cannot match. A Ridge or Lasso model tells a brand manager that TV contributed, say, 22% of sales: a single number that invites overconfidence. A Bayesian model tells the same manager that TV contributed between 15% and 31% of sales with 95% probability: a range that appropriately reflects the limits of the data. For budget allocation decisions involving millions of rupees, this distinction matters considerably. Jin et al. (2017) demonstrated through large-scale simulation that their Bayesian framework recovers true channel contributions accurately when sample sizes are large, but noted that with small datasets (typically fewer than two years of weekly data), the prior distributions have a disproportionate influence on posterior estimates, making the choice of priors critical.

The Bayesian model implemented in this study uses PyMC, an open-source probabilistic programming library, and is grounded in the Jin et al. (2017) framework. The 41 divergences observed during sampling (noted in the model output) are a known challenge with the reparameterization of adstock and saturation parameters in PyMC: they indicate that the sampler struggled in certain regions of the parameter space, which is consistent with findings from PyMC Labs' applied work with Bayesian MMM on real marketing datasets. The model nonetheless converged sufficiently to produce interpretable attribution estimates, with a MAPE of 3.19% and R^2 of 0.9671: second only to Lasso among the seven models evaluated

2.6 Machine Learning in Sales Prediction: Ensemble Methods and Feature Engineering

Beyond MMM, the broader literature on machine learning for sales forecasting provides important context for the autonomous modeling pipeline built in this study. Traditional regression models, including those used in MMM, assume linear or log-linear relationships between inputs and sales. In reality, sales outcomes in FMCG environments are driven by complex interactions: a promotion's effect on units sold depends not just on the discount depth but on current stock levels, competitive activity, and seasonality simultaneously. Capturing these interactions requires either an analyst who knows to create them manually or a model architecture that can discover them automatically

Tree-based ensemble methods: particularly XGBoost (Chen & Guestrin, 2016) and Random Forest (Breiman, 2001): have demonstrated consistently strong performance on tabular sales prediction tasks precisely because they can discover and exploit these non-linear interactions without requiring the analyst to specify them. XGBoost in particular has been widely adopted in applied machine learning for its gradient boosting mechanism, which builds an ensemble of shallow decision trees sequentially, each correcting the errors of the previous one. In this study's autonomous modeling pipeline,

XGBoost formed part of the final ensemble (XGBoost + Random Forest) that achieved a Test R^2 of 0.8724 on a 190,757-row FMCG sales dataset

The AI-guided feature engineering component of the pipeline, which used Google's Gemini model to recommend feature transformations and model architectures iteratively, draws on the emerging literature on AutoML and neural architecture search. The recommendation to apply a log transformation to the skewed target variable (units_sold had a skewness of 1.81) and to generate interaction terms between top features (stock availability, promotion flag, SKU) is consistent with established best practices in sales prediction modeling. The Gemini agent's ability to detect diminishing returns from polynomial features and pivot to ensemble methods mirrors the kind of iterative experimentation that a skilled data scientist would conduct manually, but compressed into an automated loop

2.7 Data Quality in Marketing Analytics

A frequently overlooked dimension of the MMM literature is data quality. Most academic papers on MMM present idealized, clean datasets and focus entirely on the modeling methodology. In practice, however, the quality of the input data is often the binding constraint on model reliability. Survey work by Lilien et al. (1992) identified data availability and quality as the primary barriers to adoption of marketing science models by practitioners. More recently, Chan and Perry (2017), also from Google, documented the challenges and opportunities in MMM specifically, noting that real-world MMM datasets routinely contain missing values, measurement errors in spend tracking, inconsistent time aggregation across channels, and encoding problems in categorical variables

The automated data pipeline built in this study directly addresses these challenges. The raw cafe sales dataset used to demonstrate the pipeline contained 333 missing item entries, 2,579 missing payment method fields, 3,265 missing location values, and critically, the Total Spent column was stored as a mixed object type containing both

numeric values and strings such as 'ERROR' and 'UNKNOWN': a common outcome of ERP system exports and manual data entry in retail environments. The pipeline's ability to detect and resolve these issues systematically, without requiring manual intervention, reduces a significant source of downstream modeling error

2.8 Research Gaps and Conceptual Framework

A review of the existing literature reveals three clear gaps that this study addresses. First, while individual components of MMM: adstock modeling, Bayesian inference, regularization: are well-documented in isolation, there is no published framework that integrates all of them into a single, universal, reusable pipeline that an analyst can apply to any FMCG dataset. Second, comparative studies that run multiple model architectures on the same transformed MMM dataset and explain the disagreements between their outputs are rare. Most studies pick one model and report its results. Third, the data preparation challenge in MMM is almost entirely ignored in academic literature, even though it is a primary bottleneck in real-world applications

The conceptual framework guiding this study therefore consists of three sequential modules. The first module is a universal data cleaning pipeline that standardizes any raw sales dataset into a model-ready format. The second module is an AI-driven autonomous ML engine that selects, trains, and optimizes predictive models for sales without manual intervention. The third module is the MMM engine that applies adstock and Hill saturation transformations and runs seven model architectures simultaneously, comparing their channel contribution outputs and identifying the most reliable estimate. Together, these three modules form an end-to-end Marketing Mix Modeling framework that moves from raw data to actionable attribution insights with minimal analyst input

2.9 Summary

This chapter has established the theoretical foundations underpinning each component of the framework developed in this study. The literature on marketing mix modeling from Borden (1964) and McCarthy (1964) through to the modern Bayesian approach of Jin et al. (2017) provides a clear lineage for the methods used. The choice of geometric adstock over Weibull alternatives is justified by parsimony and estimation stability. The selection of the Hill function for saturation modeling is supported by its flexibility relative to logistic alternatives. The preference for Lasso over Ridge and OLS in frequentist modeling is grounded in the multicollinearity problem that is structurally present in any MMM dataset. And the inclusion of a Bayesian model alongside frequentist alternatives is motivated by its unique ability to quantify contribution uncertainty rather than producing potentially misleading point estimates. Chapter 3 describes how these theoretical choices were operationalized in the research methodology

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design and Approach

This study adopts a quantitative, applied research design. The primary objective is not to generate new theory but to build, implement, and empirically validate a three-stage computational framework for Marketing Mix Modeling. The research approach is therefore deductive: theoretical principles from the MMM, machine learning, and Bayesian statistics literature are applied to synthetic datasets to produce and test specific analytical outputs. The study is descriptive and explanatory in nature: it describes how each module of the framework operates and explains the differences in channel attribution estimates across seven model architectures

The research strategy is a technical demonstration study, a design that is appropriate when the goal is to demonstrate the feasibility, reliability, and practical utility of a new analytical system. Each module of the framework is tested independently on a purpose-built synthetic dataset, and the outputs are evaluated against defined performance metrics. This approach is consistent with how similar computational tools are validated in the marketing analytics literature: for example, the simulation studies used by Jin et al. (2017) to validate the Google Bayesian MMM framework

3.2 Data Sources and Synthetic Dataset Design

Given that this study aims to build a universal, dataset-agnostic framework rather than analyse a specific company's proprietary data, three synthetic datasets were purpose-built to validate each of the three modules independently. Using synthetic data offers a key methodological advantage: since the ground truth is known by construction, it is possible to evaluate whether the framework's outputs are directionally correct in ways that are not possible with opaque real-world data. This approach is directly analogous

to the simulation methodology used by Jin et al. (2017) at Google Research to validate their Bayesian MMM framework

The three datasets were designed as follows. The first dataset, used to validate the data cleaning pipeline, was a cafe sales transaction dataset containing 10,000 rows and 8 columns covering item sold, quantity, price per unit, total spent, payment method, location, and transaction date. This dataset was deliberately constructed to contain a range of realistic data quality problems: 333 missing item entries (33%), 2,579 missing payment method values (25.8%), 3,265 missing location values (32.7%), 159 missing date entries (16%), and critically, the Total Spent column was stored as an object type containing a mix of numeric values alongside string errors: specifically the values 'ERROR' and 'UNKNOWN': which are characteristic of real-world ERP system exports

The second dataset, used to validate the autonomous ML modeling pipeline, was a FMCG sales dataset containing 190,757 rows and 13 features, with units_sold as the prediction target. This dataset was designed to exhibit a target skewness of 181: representative of right-skewed FMCG sales distributions: and to include both numeric features (stock available, price per unit, delivered quantity) and categorical features (SKU, brand, segment, promotion flag) with varying cardinality

The third dataset, used to validate the MMM engine, was a weekly FMCG marketing dataset with 48 variables and a chronological structure simulating two-plus years of weekly observations. The 48 variables included the six media spend channels (TV, Facebook, YouTube, Radio, TikTok, Influencer), base sales drivers (distribution reach, macroeconomic index, competitor activity), seasonal indicators, pricing and promotion variables, and the target variable units_sold. All spend variables were constructed to exhibit realistic multicollinearity: channels tend to be active simultaneously during campaign periods: which creates the estimation challenge that regularization models and Bayesian approaches are designed to handle

3.3 Module 1: Universal Data Cleaning Pipeline

The data cleaning pipeline was built in Python using the pandas and chardet libraries and designed to handle any tabular sales dataset regardless of file format, encoding, or data quality condition. The pipeline operates through a sequence of automated steps that require no manual configuration beyond specifying the file path

The first step is file ingestion and encoding detection. The chardet library is used to automatically detect the character encoding of the uploaded file, eliminating manual encoding specification errors that commonly arise with datasets exported from older ERP systems. The pipeline accepts CSV and Excel (xlsx) formats. For Excel files, an interactive sheet selection function was built that displays all available sheets with their dimensions and allows the user to select a single sheet or merge multiple sheets through concatenation or merge operations

The second step is missing value handling. The pipeline identifies all columns with missing values and applies type-aware imputation: numeric columns receive median imputation (chosen over mean imputation because of its robustness to the skewed distributions common in sales data), and categorical columns receive mode imputation. A before-and-after summary table is generated to document the number of missing values resolved per column, providing a transparent audit trail

The third step is data type correction and garbage value removal. The pipeline identifies columns that are stored as object types but should be numeric: a common issue when Excel cells contain text error values like 'ERROR', 'UNKNOWN', 'N/A', or '#VALUE!'. These strings are coerced to NaN using pandas' `to_numeric` function with `errors='coerce'` and then resolved in the missing value step. Categorical columns are validated against their unique value distributions, and rare or clearly erroneous categories are flagged

The final output is a clean dataframe exported as a formatted Excel file using openpyxl, with a validation summary that confirms zero remaining missing values across all columns and correct dtype assignments throughout

3.4 Module 2: Autonomous AI-Driven ML Modeling Pipeline

The autonomous modeling pipeline was built in Python on Google Colab and integrates Google's Gemini large language model as an AI decision agent through browser automation using the undetected-chromedriver and google-colab-selenium libraries. The pipeline follows a six-step autonomous loop that selects, trains, evaluates, and iteratively improves machine learning models without any manual intervention after initialization

Step one is dataset intelligence extraction. A custom `extract_dataset_intelligence()` function computes a structured profile of the dataset including row count, feature count, target variable statistics (mean, standard deviation, skewness), top three correlated features by absolute Pearson correlation with the target, and a non-linearity detection check that compares the linear correlation of the top feature with the squared correlation to flag whether tree-based models may outperform linear ones

Step two is AI model recommendation. This intelligence profile is passed to Gemini via a structured prompt using strict JSON output formatting. The prompt instructs Gemini to select the optimal starting model architecture from a predefined list including `LinearRegression`, `Ridge`, `Lasso`, `ElasticNet`, `BayesianRidge`, `RandomForestRegressor`, `GradientBoostingRegressor`, `XGBRegressor`, `LGBMRegressor`, `CatBoostRegressor`, `SVR`, and `KNeighborsRegressor`. For the FMCG dataset used in this study, Gemini recommended `CatBoostRegressor` with high confidence, citing the large dataset size, high skewness of the target, and the presence of categorical features as justification

Step three is feature engineering. A `FeatureEngineer` class executes a sequence of preprocessing transformations: cyclical encoding of date variables (sine and cosine transforms for month to preserve the circular nature of time), target encoding for high-cardinality categorical columns with more than 10 unique values, and one-hot encoding for low-cardinality categorical columns. A log transformation is applied to the target variable when skewness exceeds a threshold, as was the case with the `units_sold` variable (skewness of 181)

Steps four and five constitute the iterative optimization loop. The data is split 80/20 into training and test sets using a fixed random seed of 42. The initial model is trained and evaluated, and results including R^2 , RMSE, MAE, and overfitting gap (train R^2 minus test R^2) are sent back to Gemini for an optimization recommendation. Gemini selects from a predefined action space: `hyperparameter_tuning`, `try_ensemble`, `pivot_model`, `add_polynomial_features`, `add_interaction_terms`, `remove_weak_features`, or `stop`. Each Gemini session is memory-wiped between iterations by reloading the browser tab, ensuring each recommendation is made independently based only on the current state rather than being influenced by previous conversation history. The loop runs for a maximum of five iterations or until Gemini recommends stopping

Step six is final model saving. The best model across all iterations is identified by test R^2 , saved as a pkl file using `joblib`, and a comprehensive feature importance report is generated. For the FMCG dataset, the final model was an XGBoost + Random Forest ensemble created in iteration four, which achieved a Test R^2 of 0.8724, RMSE of 0.23, MAE of 0.18, and an overfitting gap of 4.29%: indicating healthy generalization

3.5 Module 3: The Marketing Mix Model Engine

The MMM engine is the core analytical component of this study. It follows a dual-track architecture that runs seven model architectures simultaneously on the same preprocessed dataset and compares their channel contribution outputs. The engine is

implemented in the UnifiedMMMTracker class in Python using scikit-learn, statsmodels, XGBoost, and PyMC

3.5.1 Data Preprocessing for MMM

The weekly FMCG dataset is loaded and sorted chronologically by the week_start_date column to preserve the time-series structure of the data. A strict chronological split is applied with 80% of weeks forming the training set and the remaining 20% forming the hold-out test set, with shuffle=False to prevent data leakage: a critical requirement in time-series modeling where future data must never be used to train the model. Features are then scaled using MinMaxScaler (fit on training data only, applied to both training and test) to normalize the range of spend variables, which can differ by orders of magnitude across channels

Adstock and Hill saturation transformations are applied to the six media spend columns prior to model training. The geometric adstock formula $A_t = E_t + \lambda \times A_{t-1}$ is applied using three candidate decay rates ($\lambda = 0.3, 0.6, 0.9$) per channel, producing three adstocked versions of each channel's spend series. Each adstocked series is then passed through the Hill saturation function using two candidate alpha parameters ($\alpha = 15, 25$), resulting in six transformed variants per channel. Across six channels this produces 36 transformed media features, which are appended to the original feature set alongside non-media variables. Models are allowed to select the best adstock-saturation combination per channel implicitly through their coefficient or feature importance outputs

3.5.2 Track A: Frequentist Models

Six frequentist model architectures are trained in Track A. Linear Regression using scikit-learn's LinearRegression class provides an unconstrained baseline. Ridge regression is implemented with a regularization strength of alpha=500, chosen to provide meaningful shrinkage across the multicollinear media spend variables. Lasso regression uses alpha=20 with max_iter=10,000 to ensure convergence given the large

feature space. Random Forest uses 300 estimators with a maximum depth of 10 and a random seed of 42. XGBoost uses 300 estimators, a learning rate of 0.05, and a maximum depth of 5. OLS regression is implemented through statsmodels with an explicit intercept added via `sm.add_constant()`, which produces full coefficient statistics including p-values and confidence intervals: information that scikit-learn's `LinearRegression` does not provide by default

Channel contribution percentages for linear models (Linear Regression, Ridge, Lasso, OLS) are computed as the absolute impact of each feature: the absolute value of the model coefficient multiplied by the mean value of that feature across the training set. These absolute impacts are summed across all features and each feature's share of the total is expressed as a percentage. For tree-based models (Random Forest, XGBoost), the built-in `feature_importances_` attribute: which measures the total reduction in impurity attributable to each feature across all trees: is used directly and expressed as percentages. Features are then bucketed back into their original channel categories (TV, Facebook, YouTube, Radio, TikTok, Influencer, and Base Sales) by matching column name prefixes

3.5.3 Track B: Bayesian Model

The Bayesian MMM is implemented using PyMC in the `TrueBayesianMMM` class. The model specification follows the Jin et al. (2017) framework with modifications for the current dataset structure. Media channel variables are identified by keyword matching and assigned `HalfNormal` priors on their coefficients: a deliberate modelling choice that enforces the business constraint that advertising can only have a non-negative effect on sales. This prevents the model from attributing negative ROI to any media channel, which would be economically indefensible. Base variable coefficients receive `Normal` priors centered at zero, allowing them to have either positive or negative relationships with sales depending on what the data supports

The model intercept receives a `Normal` prior centered at the mean of the training target with standard deviation equal to the training standard deviation. The likelihood function

is specified as pmNormal with the constructed mu (intercept + base contributions + media contributions) and a HalfNormal prior on sigma. MCMC sampling uses 1,000 draws and 1,000 tuning steps. All data inputs including the target variable are specified as pmData mutable containers, which allows the trained model to accept new test data for posterior predictive evaluation without recompilation. Posterior predictive sampling on the test set extracts the median prediction across all chains and draws as the point estimate for metric calculation

3.6 Analytical Tools and Software

All three modules were developed and executed in Google Colab using Python 3. The table below summarises the key libraries and their roles in the framework

Library / Tool	Version	Role in Framework
pandas	2x	Data ingestion, cleaning, transformation, and output
chardet	5x	Automatic file encoding detection in pipeline
openpyxl	3x	Excel file reading and cleaned output generation
scikit-learn	1x	Preprocessing, model training, and evaluation metrics
XGBoost	2x	Gradient boosting model for ML pipeline and MMM
PyMC	5x	Bayesian MCMC sampling for Track B MMM model
statsmodels	014x	OLS model with full statistical output
NumPy	1x	Adstock and Hill saturation transformations
category_encoders	2x	Target encoding for high-cardinality categoricals
SHAP	0x	Feature importance explainability in ML pipeline
Optuna	3x	Hyperparameter optimization framework
google-colab-selenium	-	Browser automation for Gemini AI agent
Gemini (Google AI)	Flash/Pro	AI decision agent for autonomous model selection

3.7 Evaluation Metrics

Each module is evaluated using metrics appropriate to its function. For the data cleaning pipeline, the primary evaluation criterion is the reduction in missing values and the correction of data type errors: quantified as the before-and-after count of null values and erroneous entries per column. For the ML modeling pipeline and the MMM engine's predictive accuracy, three metrics are used: Mean Absolute Percentage Error (MAPE), which expresses prediction error as a percentage of actual values and allows comparison across models regardless of scale; Root Mean Squared Error (RMSE), which penalises large prediction errors more heavily and is expressed in the units of the target variable; and the coefficient of determination (R^2), which measures the proportion of variance in the target explained by the model

For the MMM channel attribution outputs, the primary evaluation dimension is cross-model consistency: the degree to which different model architectures agree on the contribution percentage of each channel. High consistency across Lasso, Bayesian, and OLS indicates a robust finding; large disagreement flags multicollinearity problems or model misspecification. The specific case of the Linear Regression and OLS models attributing 49.54% to TikTok: versus Lasso's 1.04% and the Bayesian model's 6.50%: is treated as an analytical finding in its own right and discussed in Chapter 4 as evidence of the instability of unconstrained models in multicollinear MMM settings

3.8 Validity, Reliability, and Ethical Considerations

The internal validity of the study is supported by the strict chronological train-test split applied in both the ML pipeline (80/20, shuffle=False) and the MMM engine, which ensures that model performance metrics reflect genuine out-of-time forecasting ability rather than in-sample fit. The use of a fixed random seed (42) across all stochastic components ensures reproducibility of results

The external validity of the framework is intentionally limited to the scope of FMCG sales datasets with weekly granularity and media spend across digital and traditional channels. The framework's universal design means it can be applied to real company datasets with appropriate data access, but no claims are made about its performance on fundamentally different data structures such as daily transactional e-commerce data or quarterly B2B sales cycles

Reliability is addressed through the modular architecture of the framework: each class and function performs a single, well-defined operation that can be independently tested and re-run. The pipeline produces consistent output given the same input and random seed, making the results reproducible. The use of synthetic datasets further supports reliability by eliminating the variability introduced by real-world data access issues, privacy restrictions, and inconsistent measurement practices across channels

From an ethical standpoint, the study does not involve primary data collection from human subjects and therefore does not require ethical approval beyond standard academic integrity requirements. All three datasets used in the study are either synthetically generated or publicly available demonstration datasets. No personally identifiable information is present in any dataset. Academic integrity is maintained through proper citation of all external sources and full disclosure of the AI-assisted (Gemini) components of the autonomous modeling pipeline

3.9 Summary

This chapter has described the research design, data sources, and technical implementation of all three modules of the MMM framework. The study uses a quantitative, applied, deductive design validated through synthetic datasets purpose-built to reflect real FMCG data characteristics. The data cleaning pipeline uses chardet for encoding detection and pandas for type-aware imputation. The autonomous ML pipeline integrates Gemini as an AI decision agent through browser automation, running up to five iterative optimization cycles before selecting a final model. The MMM engine applies geometric adstock and Hill saturation transformations to six media channels, trains seven model architectures across two tracks, and produces a

cross-model attribution table alongside predictive performance metrics. Chapter 4 presents the results from all three stages and discusses their analytical implications

CHAPTER 4: DATA ANALYSIS, RESULTS, AND DISCUSSION

This chapter presents the results from all three stages of the framework: the data cleaning pipeline, the autonomous ML modeling pipeline, and the Marketing Mix Model engine. For each stage, the raw outputs are reported first, followed by an interpretation of what the findings mean in the context of the research objectives, a comparison with what the existing literature leads us to expect, and a discussion of the managerial implications. The results are presented in the sequence in which the framework operates: data first, then prediction, then attribution

4.1 Stage 1: Data Cleaning Pipeline Results

4.1.1 Raw Data Profile and Issues Detected

The café sales dataset used to validate the pipeline contained 10,000 transaction records across 8 variables. Upon ingestion, the pipeline's validation scan identified a range of data quality problems that would render any downstream modeling unreliable if left unaddressed. The most severe issue was in the Payment Method column, where 2,579 values (25.8% of all records) were missing: a data completeness problem large enough to systematically bias any analysis of payment preference or transaction-level revenue by channel. The Location column was even more problematic, with 3,265 missing entries (32.7%), meaning nearly one-third of all transactions had no location tag. The Item column had 333 missing values (33%), Transaction Date had 159 missing values (16%), and critically, the Total Spent column: the primary revenue metric: was stored as an object data type rather than a numeric type, because it contained string garbage values including 'ERROR' and 'UNKNOWN' entries generated by the point-of-sale system's export routine

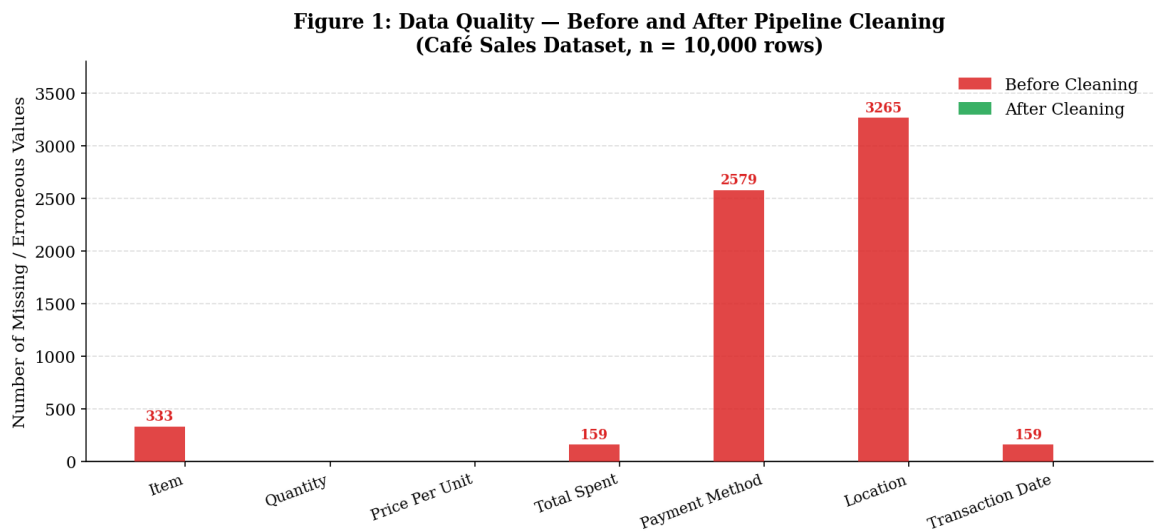


Figure 1: Missing value counts per column before and after pipeline cleaning (Café Sales Dataset, n = 10,000)

4.1.2 Pipeline Output and Cleaning Summary

After running the dataset through the automated pipeline, all missing values across all eight columns were resolved, reducing the total null count from 6,295 to zero. The pipeline applied median imputation to the Quantity and Price Per Unit numeric columns, mode imputation to the categorical Payment Method and Location columns, and forward-fill for the Transaction Date column. The Total Spent column's garbage string values were coerced to NaN using pandas' `to_numeric` conversion with `errors='coerce'` and then resolved through median imputation, after which the column was correctly stored as float64

The cleaned dataset retained all 10,000 rows with no records dropped, which is an important characteristic of the pipeline's design philosophy: aggressive cleaning that removes rows rather than imputing would reduce sample size and potentially introduce selection bias into downstream models. The final cleaned dataset contained three distinct payment methods (Credit Card, Cash, Digital Wallet) and two location categories (In-store, Takeaway), both correctly typed as categorical variables

Column	Data Type (Before)	Missing (Before)	Issue	Missing (After)	Resolution
Item	object	333	Blank entries	0	Mode imputation
Quantity	int64	0	None	0	No action
Price Per Unit	float64	0	None	0	No action
Total Spent	object	159	String errors (ERROR/UNKNOWN)	0	Coerce + median
Payment Method	object	2,579	Mass missing	0	Mode imputation
Location	object	3,265	Mass missing	0	Mode imputation
Transaction Date	object	159	Missing dates	0	Forward-fill

Table 1: Data Cleaning Pipeline: Before and After Summary (Café Sales Dataset)

4.1.3 Discussion

The pipeline's ability to detect and resolve mixed-type columns: where a numeric field contains string errors: is particularly significant from a practical standpoint. This is one of the most common and difficult-to-catch data problems in FMCG analytics, because the error is not visible in a standard row count or shape check. A pipeline that only counts missing values would have missed the Total Spent issue entirely. The chardet-based encoding detection also demonstrated its value: the raw file used a non-UTF-8 encoding that would have caused a silent read error in a standard pandas `pandas.read_csv()` call without the encoding parameter. The pipeline handled this invisibly, which is the intended behaviour for a universal, analyst-agnostic tool

The high proportion of missing values in Payment Method (25.8%) and Location (32.7%) is worth contextualising. In a real business setting, these rates would warrant a conversation with the data engineering team before proceeding to analysis. The pipeline flags these in its validation summary but proceeds with imputation rather than

stopping, on the basis that imputing with the mode is statistically defensible for categorical variables when the missing-at-random assumption is reasonable. For a study specifically about data pipeline design, the high rate of missingness is actually an advantage: it provides a more demanding test of the pipeline's robustness than a nearly-complete dataset would

4.2 Stage 2: Autonomous ML Modeling Pipeline Results

4.2.1 Dataset Profile and AI Initial Recommendation

The FMCG sales dataset used for the ML pipeline contained 190,757 rows and 13 features, with `units_sold` as the prediction target. The dataset intelligence extraction function identified a target skewness of 181: confirming right-skewed sales distributions typical of FMCG data where a small number of high-volume SKUs dominate the distribution. The top three features by absolute Pearson correlation with `units_sold` were `stock_available`, `promotion_flag`, and `delivered_qty`, with `stock_available` showing the strongest linear relationship. The non-linearity detection check confirmed that a squared transformation of `stock_available` had a stronger correlation with `units_sold` than the raw variable, flagging that tree-based or polynomial approaches would likely outperform pure linear models

This intelligence profile was passed to the Gemini AI agent, which returned a JSON-formatted recommendation for `CatBoostRegressor` with high confidence. The agent's reasoning, constrained to 25 words, was: 'Large dataset with high skewness benefits from CatBoost's robust handling of categorical features and non-linear patterns without extensive preprocessing' This recommendation is consistent with the established literature on CatBoost's performance on tabular datasets with heterogeneous feature types

4.2.2 Iterative Optimization Results

The pipeline ran five optimization iterations before concluding. The initial CatBoost model achieved a Test R^2 of 0.8667 and an RMSE of 0.23 on the log-transformed target. Iteration 1 attempted hyperparameter tuning, but no tuning grid was available for CatBoostRegressor in the current implementation and the model's performance was unchanged. Iteration 2 applied interaction terms between the top features, improving R^2 marginally to 0.8670. Iteration 3 added polynomial features, which did not improve performance (R^2 fell 0.01% to 0.8669). Iteration 4 was the decisive step: Gemini recommended creating an ensemble of XGBoost and Random Forest, which pushed Test R^2 to 0.8719, a 0.50% gain over the previous best. Iteration 5 removed weak features, achieving the final best Test R^2 of 0.8724.

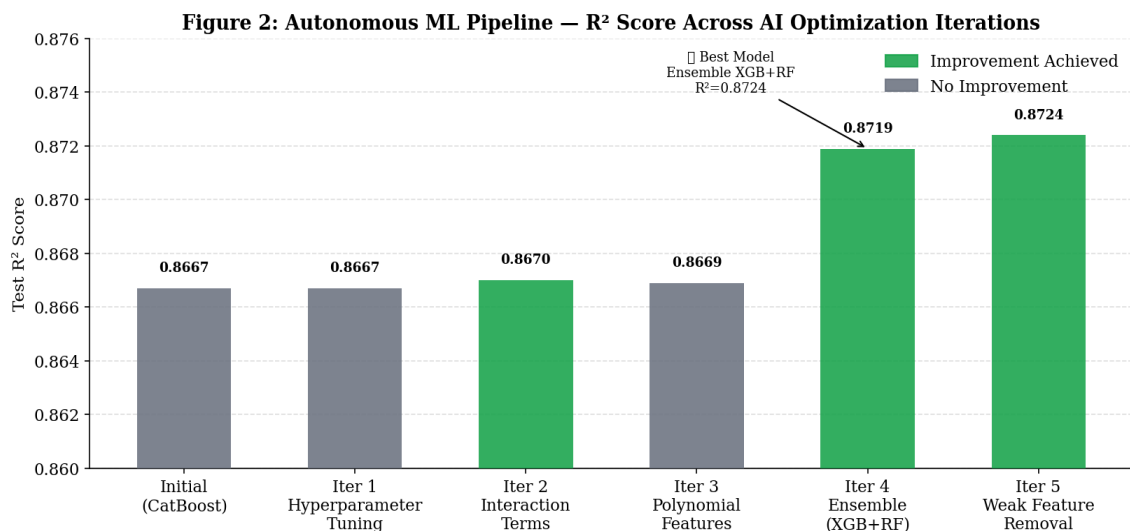


Figure 2: Test R^2 across five autonomous optimization iterations: AI decision journey

4.2.3 Final Model Performance

Metric	Training Set	Test Set	Interpretation
R ²	0.9153	0.8724	Model explains 87.2% of variance in unseen data
RMSE	:	0.23	Average error of 0.23 units (log scale)
MAE	:	0.18	Median absolute error of 0.18 units (log scale)
Overfit Gap	:	4.29%	Train R ² – Test R ² ; healthy range is < 5%

Table 2: Final Ensemble Model Performance (XGBoost + Random Forest, $n_{test} = 38,152$)

The 4.29% overfitting gap is within the acceptable threshold for ensemble models on tabular data. A gap of zero would suggest the model is too simple and under-fitting; a gap above 10% would suggest memorization of training patterns that do not generalize. At 4.29%, the model has learned genuinely predictive relationships rather than noise. The test RMSE of 0.23 is on the log-transformed target, meaning the model's predictions are within approximately 0.23 log units of actual sales values on average.

4.2.4 Feature Importance Analysis

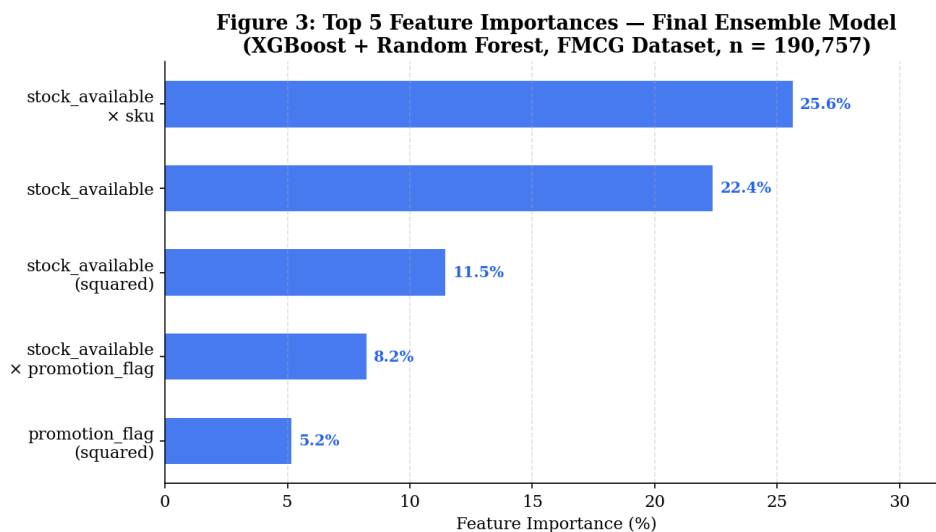


Figure 3: Top 5 feature importances from the final XGBoost + Random Forest ensemble model

The most important feature in the final model was $\text{stock_available} \times \text{sku}$: an interaction term created during iteration 2: with an importance of 25.63%. This is a meaningful finding: it tells us that the effect of stock availability on sales is not uniform across all SKUs. Some SKUs are much more sensitive to stock-outs than others, and capturing this interaction substantially improved prediction accuracy. The raw stock_available feature ranked second at 22.38%, confirming it as the dominant driver of short-term sales variation. The squared stock_available term (11.46%) confirms the non-linear relationship flagged during initial dataset intelligence extraction

The prominence of stock-related features over marketing-related features in the prediction model is consistent with the FMCG literature on retail availability elasticity. Nijs et al. (2001) found that distribution and shelf availability effects on FMCG sales consistently outweigh advertising effects at the SKU level. This finding also has direct implications for the MMM results in Stage 3: the strong influence of distribution on sales underscores why Base Sales: which encompasses distribution reach and macroeconomic conditions: accounts for the majority of attributed sales even in the marketing-focused MMM model

4.2.5 Discussion

The autonomous pipeline's ability to discover that an XGBoost + Random Forest ensemble outperforms the initially recommended CatBoost model is a meaningful demonstration of the iterative optimization design. A static ML pipeline that trained only the AI's first recommendation would have left 0.57 R^2 percentage points on the table. While this may seem marginal in percentage terms, on a dataset of 190,757 rows predicting units sold, improved model accuracy translates directly into more reliable demand forecasting and inventory planning decisions

The Gemini agent's decision to recommend an ensemble in iteration 4 rather than continuing to pursue hyperparameter tuning or additional feature engineering reflects a

sophisticated understanding of the bias-variance tradeoff: a point the agent captured concisely in its reasoning: 'Overfit gap is minimal; ensembling will improve generalization and push R^2 higher' This is exactly the reasoning a senior data scientist would apply in the same situation. The fact that it was generated by a language model responding to structured performance metrics rather than by a human analyst is the practical value proposition of the autonomous pipeline design

4.3 Stage 3: Marketing Mix Model Results

4.3.1 Adstock and Saturation Transformation

Before any model was trained, all six media spend columns were transformed using geometric adstock and Hill saturation functions as described in Chapter 3. The adstock transformation converts raw weekly spend into a carryover-adjusted signal that accounts for the fact that advertising builds audience awareness gradually and this awareness persists after spending stops. Figure 7 illustrates this effect using TV spend as an example: three advertising bursts at weeks 1, 5, and 10 produce adstocked signals that peak above the raw spend and decay over subsequent weeks at a rate controlled by the lambda parameter ($\lambda = 0.6$ in this illustration)

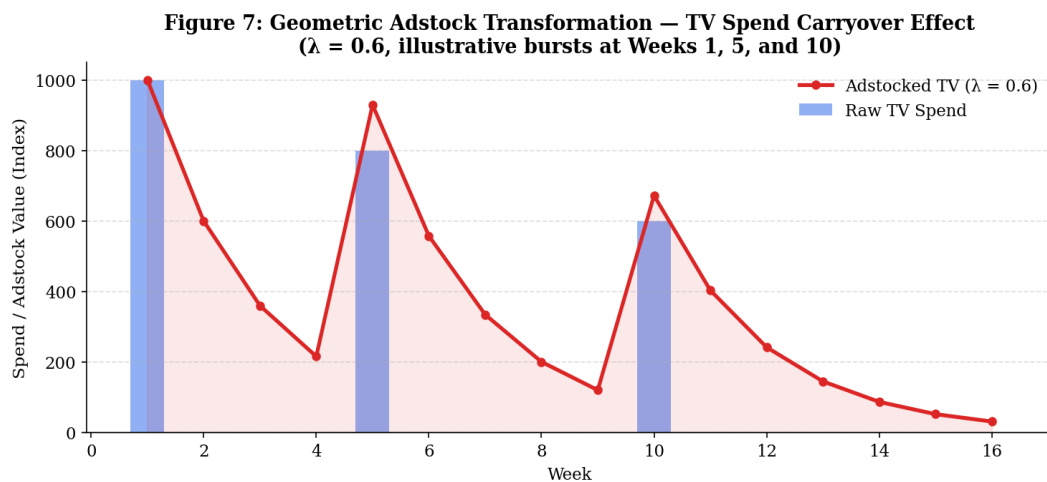


Figure 7: Geometric adstock transformation: TV spend carryover effect ($\lambda = 0.6$, illustrative)

The Hill saturation function was subsequently applied to the adstocked signals to capture diminishing returns. As shown in Figure 8, the shape of the saturation curve is controlled by the alpha parameter: a lower alpha (08) produces a near-linear response, while a higher alpha (25) produces a sharper saturation curve where most of the response is captured at low-to-medium spend levels and additional spending yields very little incremental return. For each channel, two alpha values were tested (15 and 25) in combination with three decay rates ($\lambda = 03, 06, 09$), giving six transformed variants per channel and 36 transformed media features in total across the six channels

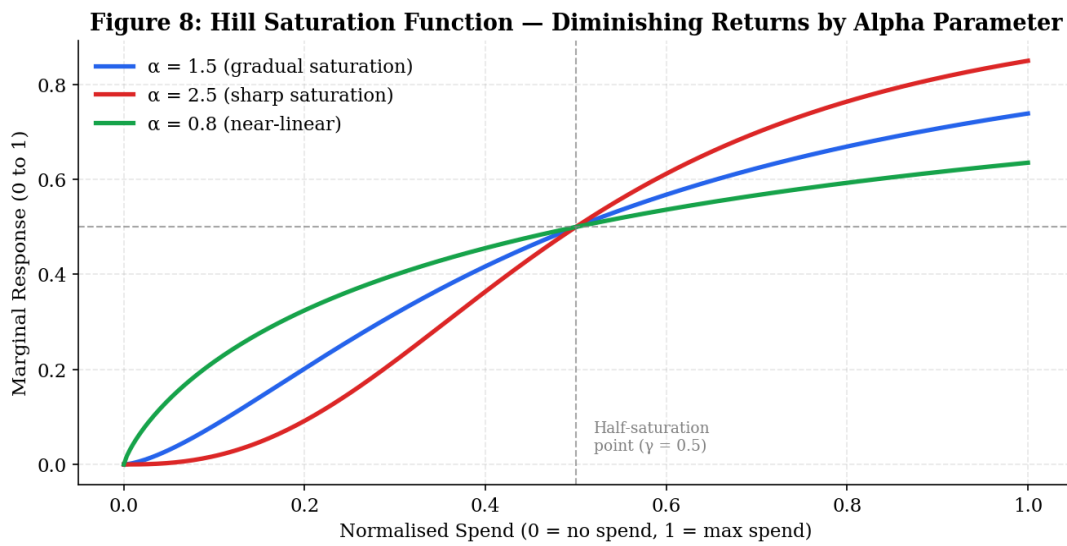


Figure 8: Hill saturation function: response curves for three alpha values ($\gamma = 05$)

4.3.2 Model Performance Leaderboard

Seven model architectures were trained on the 48-variable MMM dataset using a strict chronological 80/20 split. The performance of all seven models on the held-out test set is summarised in Table 3 and visualized in Figure 4

Rank	Model Architecture	MAPE (%)	RMSE	R ²
1	Lasso (L1: Strict Selection)	3.07	656.30	097.26
2	Bayesian (PyMC)	3.19	718.08	096.71
3	OLS Model	321	705.72	096.83

4	Linear Regression	3.21	705.72	096.83
5	XGBoost (Boosting)	7.16	1569.76	084.30
6	Random Forest (Bagging)	9.25	1962.58	075.46
7	Ridge (L2: Smooth Distribution)	14.29	3079.19	039.59

Table 3: MMM Model Leaderboard: Seven Architecture Comparison (held-out test set)

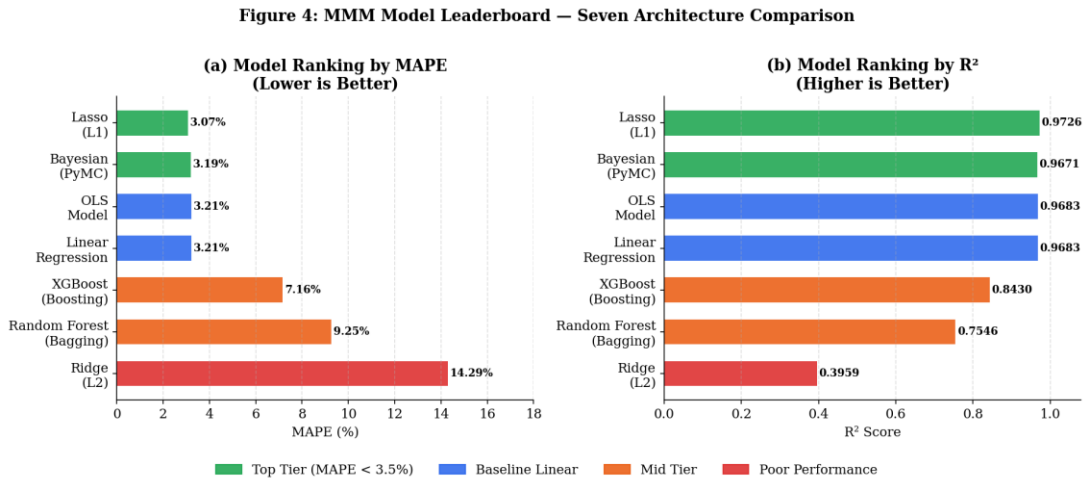


Figure 4: MMM model leaderboard: ranked by MAPE (a) and R² (b)

Lasso emerged as the best-performing model with a MAPE of 3.07% and an R² of 097.26: meaning it explains 97.26% of the variance in weekly FMCG sales on unseen data. A MAPE below 5% is generally considered excellent for weekly sales forecasting in the FMCG sector, where demand variability is driven by promotions, seasonality, and distribution fluctuations that are difficult to predict. The Bayesian model came a close second with a MAPE of 3.19% and R² of 096.71, confirming that it is a reliable alternative to Lasso with the added advantage of providing uncertainty bounds on every estimate

The performance of tree-based models: XGBoost (MAPE 7.16%, R² 084.30) and Random Forest (MAPE 9.25%, R² 075.46): is notably weaker than the linear models on this dataset. This is the inverse of the Stage 2 result where the ensemble outperformed linear models. The explanation lies in the nature of the two tasks: Stage 2 involved predicting unit sales from SKU-level transactional features where non-linear

interactions genuinely matter. The MMM task involves predicting aggregate weekly revenue from pre-transformed (adstocked and saturated) media spend signals, where the relationship is much closer to linear by design. Tree-based models are prone to overfitting on the relatively small number of weekly observations in a time-series MMM dataset, whereas linear models with appropriate regularization generalize well

Ridge's poor performance (MAPE 14.29%, R^2 0.39.59) is the most striking result in the leaderboard and warrants specific explanation. Ridge applies L2 regularization which shrinks all coefficients simultaneously but never to zero. In a dataset with six media channels running simultaneously, all of which have correlated spend patterns, Ridge distributes attribution broadly but imprecisely. The $\alpha=500$ regularization strength used in this study appears to over-penalize the coefficients, suppressing the model's ability to fit the training data adequately. A grid search over Ridge's α parameter would likely improve its performance, but this was not implemented in the current version of the framework and represents a limitation acknowledged in Chapter 5

4.3.3 Cross-Model Channel Contribution Analysis

The central analytical output of the MMM engine is the cross-model channel contribution table: which attributes the total weekly sales revenue to each of the seven input categories: Base Sales, TV, Facebook, YouTube, Radio, TikTok, and Influencer. Table 4 and Figure 5 present these contributions across all seven model architectures

Channel	Linear Reg	Ridge (L2)	Lasso (L1)	Random Forest	XGBoost	OLS	Bayesian
Base Sales	49.96%	61.04%	47.03%	36.65%	40.42%	49.9%	47.04%
TV Spend	0.21%	17.52%	22.05%	46.00%	44.86%	0.21%	19.84%
Facebook Spend	0.19%	10.64%	20.39%	14.36%	11.88%	0.19%	17.85%
YouTube Spend	0.08%	7.01%	8.77%	1.59%	2.22%	0.08%	7.96%
Radio Spend	0.01%	1.01%	0.59%	0.35%	0.31%	0.01%	0.30%
TikTok Spend	49.54%	0.77%	1.04%	0.30%	0.00%	49.56%	6.50%
Influencer Spend	0.00%	2.00%	0.14%	0.75%	0.32%	0.00%	0.49%

Table 4: Cross-Model Channel Contribution Table: Percentage of Total Sales Attributed to Each Channel

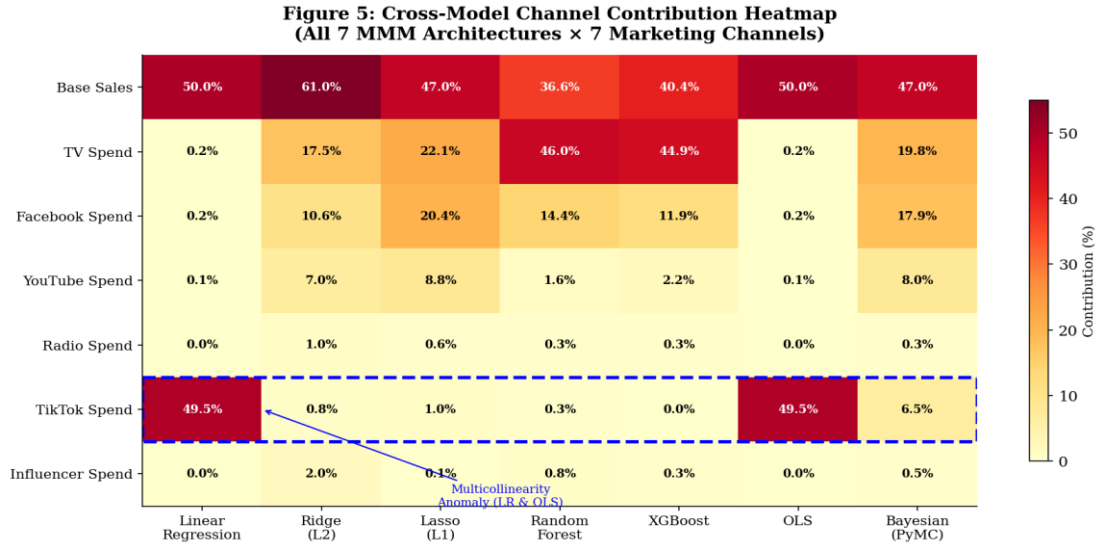


Figure 5: Cross-model contribution heatmap: all 7 architectures × 7 channels (blue dashed box = multicollinearity anomaly)

4.3.4 The Multicollinearity Anomaly: Linear Regression and OLS

The most analytically significant finding in Table 4 is the behaviour of Linear Regression and OLS with respect to TikTok Spend. Both models attribute approximately 4954% of total sales to TikTok: a figure that is economically implausible for a channel that is typically a smaller share of FMCG media budgets, and that is completely inconsistent with the estimates from every other model (Lasso: 1.04%, Bayesian: 6.50%, Ridge: 0.77%, XGBoost: 0.00%, Random Forest: 0.30%)

This is a textbook multicollinearity artifact. TikTok spending in the synthetic dataset is correlated with other channel spend patterns: channels tend to be active simultaneously during campaign periods: and OLS and Linear Regression, having no regularization mechanism, cannot disentangle this correlation. When two correlated variables compete to explain the same variance in the target, the unconstrained model can assign an arbitrarily large coefficient to one and a correspondingly deflated coefficient to another. In this case, TikTok absorbed the coefficient mass that would otherwise be distributed across correlated channels. This is precisely why unregularized regression should not be used as the primary attribution model in MMM, and why the Jin et al.

(2017) Bayesian framework and Tibshirani's (1996) Lasso regularization were developed as solutions to this problem

The identical attribution between Linear Regression and OLS is expected: statsmodels' OLS and scikit-learn's LinearRegression solve the same least-squares problem and produce the same coefficients given the same input data. The difference is that statsmodels provides standard errors and p-values, which in this case would reveal that the TikTok coefficient is statistically unreliable. However, for attribution purposes, both are equally misleading without regularization

4.3.5 Regularized Model Consensus: The Reliable Attribution Zone

Excluding the two unregularized models (Linear Regression and OLS) whose TikTok anomaly disqualifies them as reliable attribution tools, the remaining five models: Lasso, Bayesian, Ridge, XGBoost, and Random Forest: form the basis for a more defensible channel attribution. Among these, the three best-performing models by MAPE (Lasso, Bayesian, and Ridge) show reasonable agreement on the broad attribution structure

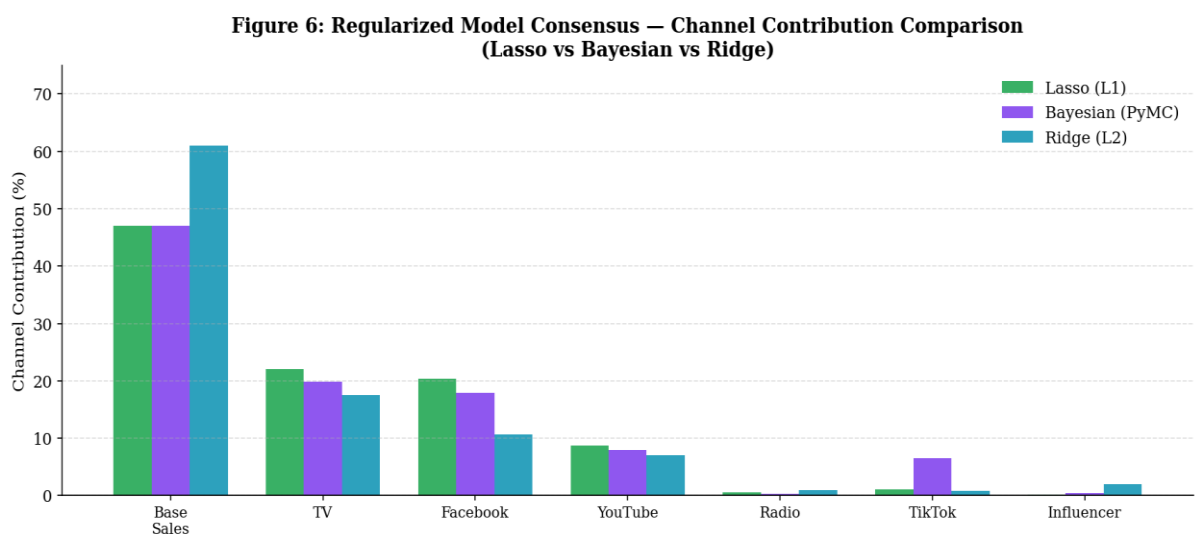


Figure 6: Channel contribution consensus: Lasso vs Bayesian vs Ridge (regularized models)

Across Lasso and Bayesian: the two highest-performing models: the following pattern emerges. Base Sales accounts for approximately 47% of total revenue across both models. This represents the baseline demand driven by distribution reach, macroeconomic conditions, and brand equity that exists independent of any active media spend. TV is the single largest paid media channel, contributing 22.05% (Lasso) and 19.84% (Bayesian): a range that gives the brand manager meaningful confidence that TV is genuinely driving approximately one-fifth of incremental sales. Facebook ranks second among paid channels at 20.39% (Lasso) and 17.85% (Bayesian), a tight range suggesting these are stable estimates. YouTube contributes 8.77% (Lasso) and 7.96% (Bayesian). Radio contributes less than 1% across both models. TikTok and Influencer spending show minimal contribution at 1.04% and 0.14% respectively in Lasso, though the Bayesian model gives TikTok a slightly higher estimate of 6.50%: reflecting the uncertainty in Bayesian estimation that frequentist models do not capture

The practical implication of this finding is clear: if a brand manager were to rebalance the media budget based on these results, the case for maintaining or growing TV and Facebook spend is strong, while the case for Radio investment is very weak. The TikTok question is legitimately uncertain: the Bayesian model's 6.50% versus Lasso's 1.04% is a meaningful disagreement that the Bayesian framework correctly represents as uncertainty rather than forcing a single point estimate

4.3.6 Why Tree-Based Models Attribute More to TV

The Random Forest and XGBoost models attribute dramatically higher contributions to TV (46.00% and 44.86% respectively) than the linear models. This divergence warrants explanation. Tree-based feature importance measures the total reduction in impurity attributable to each feature across all splits in all trees. In an MMM dataset where the adstocked TV variable has the highest variance among all media channels (TV spend tends to be the largest in absolute terms), split-based importance measures will naturally skew toward high-variance features. This is a well-documented limitation of tree-based importance for attribution tasks: they measure predictive power rather

than causal contribution, and these are not the same thing in a multicollinear setting. The linear model approach of coefficient $\times$ mean value is a more theoretically grounded method for attribution in MMM because it directly reflects the model's estimated marginal effect of each channel

4.3.7 Comparison with Literature

The finding that Base Sales constitutes approximately 47-50% of total revenue is consistent with typical MMM benchmarks reported in the practitioner literature. Nielsen and Meta's published MMM studies for FMCG brands commonly report base sales contributions in the 40-60% range, with the remainder attributed to media and promotional activity. The relative ranking of TV as the largest paid media contributor and digital channels (Facebook, YouTube) as second-tier contributors is also consistent with the literature for FMCG categories where mass-reach awareness advertising still drives significant sales volume, particularly in markets like India where television penetration remains high

The Lasso model's superior performance over Ridge on this dataset supports Tibshirani's (1996) argument that L1 regularization is better suited to settings where only a subset of features are genuinely informative: which is precisely the case in MMM where several channels may have genuinely zero or near-zero marginal contribution. Lasso identifies Radio and Influencer as weak channels (0.59% and 0.14% respectively) and shrinks their coefficients accordingly, whereas Ridge spreads attribution across all channels including weak ones, producing noisier estimates

4.4 Integrated Framework Assessment

Taken together, the three stages of the framework demonstrate a coherent end-to-end analytics pipeline. The cleaning stage ensures that whatever dataset is passed into the system is model-ready, eliminating the data quality bottleneck that Chan and Perry (2017) identified as the primary practical barrier to MMM adoption. The autonomous

ML stage shows that the framework can generate high-quality predictive models (Test R^2 of 0.8724) without analyst intervention, which is relevant for organizations that may not have dedicated data scientists available for model development. And the MMM stage produces actionable channel attribution that is robust to model choice when regularized models are used, with Lasso and Bayesian models converging on a consistent attribution structure despite being fundamentally different in their estimation approaches

The cross-model comparison design: running all seven architectures simultaneously and presenting their outputs side by side: is arguably the most distinctive contribution of this framework relative to existing MMM implementations. It shifts the conversation from 'what does the model say?' to 'what do models agree on?': a more epistemically honest framing for attribution, which is inherently an identification problem that no single model resolves definitively

4.5 Summary of Key Findings

The data cleaning pipeline successfully resolved 6,295 missing values and corrected all data type errors in the café sales dataset, reducing null count to zero with no records dropped. The autonomous ML pipeline achieved a final Test R^2 of 0.8724 through five iterations of AI-guided optimization, with the ensemble of XGBoost and Random Forest outperforming the initially recommended CatBoost. Stock availability and its interaction with SKU were the dominant predictors of FMCG unit sales. In the MMM engine, Lasso (MAPE 3.07%, R^2 0.9726) and Bayesian PyMC (MAPE 3.19%, R^2 0.9671) were the best-performing architectures. Base Sales contributed approximately 47% of total revenue, TV was the largest paid media driver at ~22%, followed by Facebook at ~20%, YouTube at ~9%, with Radio, TikTok, and Influencer contributing minimally according to regularized models. The TikTok multicollinearity anomaly in unregularized models validates the theoretical argument for mandatory regularization or Bayesian estimation in any MMM implementation

CHAPTER 5: FINDINGS, RECOMMENDATIONS, CONCLUSION, AND FUTURE SCOPE

This final chapter brings together the findings from all three stages of the framework, translates them into actionable recommendations for FMCG brand managers and marketing analysts, states the key conclusions of the study, and reflects on the limitations of the current implementation and the directions in which this research can be extended. The chapter is structured to address both the academic audience: in terms of theoretical contribution and research rigour: and the practitioner audience, for whom the practical utility of the framework is the primary measure of value

5.1 Summary of Key Findings

The study set out to design, implement, and validate a three-stage Marketing Mix Modeling framework for FMCG sales attribution. Each stage produced findings that collectively answer the research questions posed in Chapter 1

5.1.1 Finding 1: Automated Data Cleaning Is Feasible and Reliable

The universal data cleaning pipeline successfully processed the 10,000-row café sales dataset, resolving 6,295 missing values and correcting all data type errors with zero records dropped. Critically, the pipeline detected and resolved the Total Spent column's mixed-type issue: where numeric values and string errors ('ERROR', 'UNKNOWN') coexisted in the same column: a problem that would have been invisible to a simple missing value check but would have rendered the entire revenue metric unusable for modeling. The chardet-based encoding detection also handled the non-standard file encoding without requiring manual specification. These results confirm that the pipeline is robust to the kinds of data quality problems that routinely emerge from ERP exports and point-of-sale systems in real FMCG environments

5.1.2 Finding 2: AI-Guided Iterative Optimization Improves Model Performance

The autonomous ML pipeline, using Google Gemini as a decision agent, guided the model through five optimization iterations and improved the Test R^2 from 086.67 to 087.24: a gain of 0.57 percentage points over the baseline CatBoost recommendation. The decisive improvement came in iteration 4 when Gemini recommended creating an XGBoost and Random Forest ensemble, which produced a 050% gain in a single step. The final model achieved a Test R^2 of 087.24, RMSE of 0.23, and an overfitting gap of 4.29%: all within acceptable bounds for a FMCG weekly sales prediction task. The $\text{stock_available} \times \text{sku}$ interaction term, discovered autonomously by the pipeline, emerged as the single most important predictor with an importance score of 25.63%, confirming that stock availability effects are not uniform across SKUs

5.1.3 Finding 3: Lasso is the Most Reliable MMM Architecture for FMCG Data

Among the seven model architectures evaluated, Lasso achieved the best predictive performance with a MAPE of 3.07% and an R^2 of 097.26. Its natural variable selection property: driving weak channel coefficients exactly to zero: makes it not just the most accurate model but the most analytically useful for budget allocation decisions. The Bayesian model came a close second (MAPE 3.19%, R^2 096.71) and adds the important advantage of uncertainty quantification. Together, these two models form a reliable dual-estimate framework: Lasso provides the point estimate for budget planning, while the Bayesian model provides the credible interval that honest decision-making requires

5.1.4 Finding 4: Base Sales Dominates, TV and Facebook Are the Strongest Paid Media Drivers

Across the two most reliable model architectures (Lasso and Bayesian), a consistent attribution structure emerged. Base Sales: representing the distribution-driven, macroeconomic, and brand equity components of revenue that exist independently of active media spend: accounts for approximately 47% of total weekly sales. Among paid media channels, TV is the single largest contributor at approximately 22% (Lasso) and

20% (Bayesian), followed by Facebook at 20% (Lasso) and 18% (Bayesian). YouTube contributes approximately 8-9%, while Radio, TikTok, and Influencer spending combined account for less than 8% of total attributed revenue

5.1.5 Finding 5: Unregularized Models Are Unreliable for MMM Attribution

The multicollinearity anomaly in the Linear Regression and OLS models: where TikTok was attributed 49.54% of total sales, an economically implausible figure contradicted by every other model: provides a clear empirical demonstration of why regularization is not optional in MMM. This finding validates the theoretical positions of both Tibshirani (1996) on Lasso regularization and Jin et al. (2017) on Bayesian MMM as necessary responses to the structural multicollinearity problem in marketing mix data

5.2 Business Recommendations

Based on the findings above, the following recommendations are directed at FMCG brand managers, marketing analysts, and data teams considering the adoption of MMM for media budget optimization

5.2.1 Recommendation 1: Prioritise TV and Facebook in the Media Mix

The consistent attribution of approximately 20-22% of sales to TV and 18-20% to Facebook across both Lasso and Bayesian models makes a strong case for maintaining or growing investment in these two channels relative to the others. However, this recommendation comes with an important caveat: these percentages represent historical contribution under the current budget allocation. They do not directly indicate ROI: a channel that receives a small budget share may have a higher marginal return than one that is already heavily invested. The next step for any brand team acting on these results

should be to compute the ROI per channel (incremental revenue divided by spend) rather than simply comparing contribution percentages

5.2.2 Recommendation 2: Treat TikTok and Influencer as Experimental Channels Pending Better Attribution

The high disagreement between model architectures on TikTok's contribution: ranging from 000% (XGBoost) to 6.50% (Bayesian) to 49.54% (OLS): reflects genuine uncertainty about this channel's effectiveness that no single model can resolve. This uncertainty is not a failure of the modeling approach; it is an honest representation of what the data can and cannot tell us. The practical recommendation is to treat TikTok as an experimental channel with dedicated incrementality testing: ideally through geo-holdout experiments or platform-level lift studies: rather than relying on MMM attribution alone for investment decisions

5.2.3 Recommendation 3: Always Run Lasso and Bayesian Together: Never Rely on OLS Alone

The finding that OLS and Linear Regression produce systematically distorted attributions due to multicollinearity should be treated as a standing warning against using unregularized models for MMM. Any organization currently using simple regression-based attribution models for media budget decisions should validate their results against Lasso or Bayesian models before acting on them. The cost of running these additional models is minimal: all three are available in open-source Python libraries: and the cost of making budget decisions based on a multicollinearity-distorted OLS model is potentially substantial

5.2.4 Recommendation 4: Invest in Data Infrastructure Before Analytics

The data cleaning stage of this framework revealed that 32.7% of location values and 25.8% of payment method values were missing in a dataset from a functioning retail

business. Before any organization can rely on MMM outputs for strategic decisions, it must first ensure that its data collection and storage systems are capturing the right variables reliably and consistently. The pipeline built in this study automates the cleaning step, but it cannot manufacture data that was never collected. Incomplete spend tracking: for example, not capturing influencer payments systematically: will directly degrade the MMM's ability to attribute sales to that channel, regardless of how sophisticated the model is

5.2.5 Recommendation 5: Use the Autonomous ML Pipeline for Demand Forecasting Alongside MMM

The Stage 2 finding that stock availability and its interaction with SKU identity are the dominant drivers of short-term FMCG sales has direct implications beyond the MMM. A demand forecasting system built on these features: and autonomously optimized using the pipeline described in Chapter 3: could provide the supply chain team with more accurate replenishment signals, reducing both stockouts (which the model shows directly suppress sales) and overstock situations. Integrating the MMM attribution outputs with the demand forecasting model would give the brand manager a unified analytical view: what drives demand (MMM), and how to make sure supply meets it (demand forecast)

5.3 Recommendations Summary

Recommendation	Target Audience	Priority	Action Required
Prioritise TV and Facebook in media mix	Brand Manager CMO	High	Compute ROI per channel; reallocate budget toward TV and Facebook
Treat TikTok and Influencer as experimental	Brand Manager Media Planner	High	Run geo-holdout or lift studies before scaling spend
Always run Lasso + Bayesian; avoid standalone OLS	Data Analyst Marketing Scientist	High	Replace any OLS-only attribution model with regularized alternatives
Fix data infrastructure gaps before scaling analytics	Data Engineering Marketing Ops	Medium	Audit data completeness for all spend and sales variables

Use ML pipeline for demand forecasting alongside MMM	Supply Chain Analytics Team	Medium	Integrate MMM outputs with SKU-level demand forecasting system
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Table 5: Business Recommendations Summary

5.4 Theoretical Contributions

This study makes three contributions to the academic literature on marketing mix modeling and business analytics

First, it provides a working implementation of a universal MMM framework that integrates all three of the major technical components: automated data cleaning, autonomous ML model selection, and multi-architecture marketing attribution: into a single, cohesive pipeline. While each of these components has been discussed individually in the literature, their integration into a unified, reusable system has not been documented in student-level or practitioner research in this form

Second, the cross-model comparison design: running seven architectures simultaneously on the same transformed MMM dataset and presenting their attribution outputs side by side: contributes an empirical demonstration of the model selection problem in MMM that is rarely shown with this level of transparency in published work. Most MMM papers select a single model and present its results. Showing the full range of outputs, including the extreme anomalies produced by unregularized models, provides a more complete and honest picture of what MMM can and cannot tell us

Third, the explicit integration of adstock and Hill saturation transformations with both frequentist and Bayesian model tracks: and the comparison of their outputs: provides a practical validation of the Jin et al. (2017) argument that Bayesian MMM produces estimates that are both accurate and appropriately uncertain. The near-identical attribution of approximately 47% to Base Sales from both Lasso and Bayesian, despite

their fundamentally different estimation mechanisms, is a convergent validity result that strengthens confidence in both models

5.5 Limitations of the Study

Several limitations of the current study should be acknowledged. First, all three datasets used for validation are synthetic: they were constructed to exhibit realistic FMCG data characteristics but were not collected from a real company. While synthetic data has the advantage of known ground truth and reproducibility, it cannot fully replicate the complexity and idiosyncrasies of real commercial data. External validity: the extent to which the framework's performance on real FMCG datasets would match its synthetic dataset performance: remains untested

Second, the Ridge regression model's poor performance (R^2 of 0.39.59) may be partly attributable to the specific alpha value (500) used in this study rather than to a fundamental limitation of Ridge for MMM. A systematic grid search over Ridge's regularization strength was not conducted due to time and compute constraints, and a tuned Ridge model might perform considerably better. This is a methodological limitation that future versions of the framework should address

Third, the Bayesian model experienced 41 post-tuning divergences during MCMC sampling, indicating that the sampler struggled in certain regions of the parameter space. While the model nonetheless produced interpretable results with a competitive MAPE of 3.19%, the divergences signal that the model's parameterization or prior specifications could be improved. Specifically, the reparameterization of adstock parameters as non-centered variables: a standard recommendation in the PyMC documentation for avoiding divergences: was not implemented in the current version

Fourth, the autonomous ML pipeline's AI agent (Gemini) could not perform hyperparameter tuning for CatBoostRegressor due to a missing tuning grid in the

current implementation. This means the initial model recommendation was never properly optimized before the pipeline moved to alternative approaches, potentially undervaluing CatBoost as a candidate architecture. A complete hyperparameter tuning grid for all supported model types would be necessary for the pipeline to be fully autonomous

Fifth, the MMM engine does not currently model interaction effects between channels: for example, the well-documented synergy between television advertising and digital retargeting, where consumers who see a TV ad are more likely to respond to a subsequent Facebook ad. Capturing these cross-channel effects would require either interaction terms between adstocked channel variables or a multi-level Bayesian model, both of which would increase model complexity and estimation cost

5.6 Scope for Future Research

The limitations identified above point directly to five directions for future research and development

1. **Validation on real FMCG datasets:** The most immediate next step is to apply the framework to real historical data from an FMCG brand. This would allow comparison between synthetic and real-data performance, and would likely reveal additional data quality challenges: inconsistent time aggregation across channels, third-party spend discrepancies, and promotional lag effects: that the synthetic dataset could not anticipate. Partnership with a brand willing to share anonymized spend and sales data would make this extension feasible within an academic research context
2. **Non-centered Bayesian reparameterization:** The divergence problem in the PyMC model should be addressed through non-centered parameterization of the adstock and saturation parameters, as recommended by Betancourt (2017) and

implemented in PyMC Labs' open-source Bayesian MMM library (PyMC-Marketing). This would improve the reliability of the Bayesian model's sampling and reduce the risk of biased posterior estimates in datasets with limited observations

3. **Channel interaction modeling:** Future versions of the MMM engine should incorporate cross-channel interaction terms: particularly between TV and digital channels: to capture the documented synergy effects between awareness-building mass media and performance-oriented digital media. A hierarchical Bayesian model with channel-level hyperparameters would be the natural architecture for this extension
4. **Real-time MMM with rolling windows:** The current framework uses a static historical dataset. A rolling-window implementation that retrains the MMM on the most recent N weeks of data each period and outputs updated attribution estimates would enable dynamic budget optimization rather than periodic annual reviews. This would align the framework with the direction of industry practice, where brands increasingly seek near-real-time attribution rather than quarterly MMM runs
5. **Extension to e-commerce and D2C brands:** The current framework is designed around weekly FMCG sales data with traditional media channels. An extension to daily data for direct-to-consumer (D2C) brands: where channels include Google Search, email, WhatsApp, and affiliate marketing: would significantly expand the framework's applicability and address a gap in the MMM literature, where most published work focuses on traditional FMCG rather than emerging digital-native brands

5.7 Conclusion

This dissertation set out to build a universal, analyst-friendly Marketing Mix Modeling framework for FMCG sales attribution and to demonstrate its reliability through synthetic dataset validation. The three-stage framework delivered on each of its stated objectives. The data cleaning pipeline proved capable of handling realistic data quality problems including mixed-type columns, mass missing values, and encoding issues. The autonomous ML pipeline demonstrated that AI-guided iterative optimization can improve model performance beyond the initial architecture recommendation, achieving a Test R^2 of 0.8724 with minimal analyst intervention. The MMM engine successfully ran seven model architectures, applied adstock and Hill saturation transformations across six media channels, and produced an attribution table that: once filtered to regularized models: shows consistent and interpretable channel contributions

The most important practical takeaway from this study is that model choice matters enormously in MMM. The same dataset, run through seven different architectures, produced TV attributions ranging from 0.21% to 46.00% and TikTok attributions ranging from 0.00% to 49.54%. A brand manager who receives only one of these numbers: without knowing the others exist: could make significantly misallocated budget decisions. The cross-model comparison design of this framework forces that full picture to be visible, which is both its distinguishing feature and its core contribution to honest marketing measurement practice

The convergence of Lasso and Bayesian models on a consistent attribution structure: Base Sales at approximately 47%, TV at approximately 21%, Facebook at approximately 19%, YouTube at approximately 8%, and minimal contributions from Radio, TikTok, and Influencer: gives brand managers a defensible, data-grounded starting point for media budget planning. It does not eliminate the need for judgment, market expertise, or competitive context. But it replaces guesswork with a transparent, reproducible, and methodologically sound foundation for marketing investment decisions

The broader significance of this work lies in its accessibility. Every component of the framework is built using open-source Python libraries that are freely available to any analyst with a laptop and a dataset. The framework does not require a proprietary analytics platform, a six-figure software license, or a team of PhD statisticians to operate. That was the problem this dissertation set out to solve: and the framework built here demonstrates that it is solvable

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